# Exam Formula Sheet

## Epi 202: Probability

$$\begin{aligned}\mathrm{Var}(\underset{\sim}{a}\cdot\underset{\sim}{X}) &= \mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right)\\ &= \underset{\sim}{a}^{\top}\mathrm{Var}(\underset{\sim}{X})\underset{\sim}{a}\\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \mathrm{Cov}(X_i, X_j)\end{aligned}$$

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$$

$$\mathbb{E}[Y \mid Z] = \mathbb{E}[\mathbb{E}[Y \mid X, Z] \mid Z]$$

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$$

$$\mathrm{Cov}(Y, Z) = \mathbb{E}[\mathrm{Cov}(Y, Z \mid X)] + \mathrm{Cov}(\mathbb{E}[Y \mid X], \mathbb{E}[Z \mid X])$$

## Epi 203: Statistical inference

$$\mathcal{L}(\theta) \stackrel{\text{def}}{=} \mathrm{p}(\underset{\sim}{X} = \underset{\sim}{x} | \Theta = \theta)$$

$$\ell \stackrel{\text{def}}{=} \log\{\mathcal{L}(\underset{\sim}{x}|\theta)\}$$

$$\ell' \stackrel{\text{def}}{=} \frac{\partial}{\partial\theta}\ell(\underset{\sim}{x}|\theta)$$

$$\ell'' \stackrel{\text{def}}{=} \frac{\partial}{\partial\underset{\sim}{\theta}}\frac{\partial}{\partial\underset{\sim}{\theta}^{\top}}\ell(\underset{\sim}{x}|\underset{\sim}{\theta})$$

$$\ell''_{ij} = \frac{\partial}{\partial\theta_i}\frac{\partial}{\partial\theta_j}\ell(\underset{\sim}{X} = \underset{\sim}{x}|\underset{\sim}{\theta})$$

$$I \stackrel{\text{def}}{=} -\ell''(\underset{\sim}{x}|\underset{\sim}{\theta})$$

$$\mathcal{I} \stackrel{\text{def}}{=} \mathbb{E}[I(\underset{\sim}{x}|\theta)]$$

$$\hat{\theta}_{ML} \mathrel{\dot\sim} \mathrm{N}\Big(\theta, [\mathcal{I}(\underset{\sim}{\theta})]^{-1}\Big)$$

For one parameter $\theta_k$:

$$\mathrm{CI}_{1-\alpha}(\theta_k) = \Big[\hat{\theta}_k \pm z_{1-\frac{\alpha}{2}}\widehat{\mathrm{SE}}\big(\hat{\theta}_k\big)\Big]$$

$$Z_k \stackrel{\text{def}}{=} \frac{\hat{\theta}_k - \theta_{k,0}}{\widehat{\mathrm{SE}}\big(\hat{\theta}_k\big)} \mathrel{\dot\sim} \mathrm{N}(0,1) \qquad \text{under } H_0 : \theta_k = \theta_{k,0} \qquad z_k = \text{observed } Z_k$$

$$\begin{aligned}p\text{-value} &= 2\Pr(|Z| \geq |z_k|), \quad Z \sim \mathrm{N}(0,1)\\ &= 2(1 - \Phi(|z_k|))\end{aligned}$$

## Sta 108: Linear regression

$$t_k \stackrel{\text{def}}{=} \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{SE}}\big(\hat{\beta}_k\big)} \mathrel{\dot\sim} t_{n-p} \qquad \text{under } H_0 : \beta_k = \beta_{k,0} \qquad t_k^{\text{obs}} = \text{observed } t_k$$

$$\text{CI}_{1-\alpha}(\beta_k) = \Big[\hat{\beta}_k \pm t_{n-p}\Big(1 - \frac{\alpha}{2}\Big)\widehat{\text{SE}}\big(\hat{\beta}_k\big)\Big]$$

Let $T_{n-p} \sim t_{n-p}$.

$$\begin{aligned} p\text{-value} &= 2\Pr\big(|T_{n-p}| \geq |t_k^{\text{obs}}|\big) \\ &= 2\Big(1 - F_{t_{n-p}}\big(|t_k^{\text{obs}}|\big)\Big) \end{aligned}$$

$$\text{CI}_{1-\alpha}(\mu(\underset{\sim}{x}^*)) = \Big[\hat{\mu}(\underset{\sim}{x}^*) \pm t_{n-p}\Big(1 - \frac{\alpha}{2}\Big)\widehat{\text{SE}}(\hat{\mu}(\underset{\sim}{x}^*))\Big]$$

$Y^*$ denotes a new observation (not in the training data), with corresponding covariate pattern $\underset{\sim}{x}^*$. Let $\hat{Y}^* \stackrel{\text{def}}{=} \hat{\mu}(\underset{\sim}{x}^*)$.

$$\text{PI}_{1-\alpha}(Y^*|\underset{\sim}{x}^*) = \Big[\hat{Y}^* \pm t_{n-p}\Big(1 - \frac{\alpha}{2}\Big)\widehat{\text{SE}}\big(Y^* - \hat{Y}^*\big)\Big]$$

$$\widehat{\text{SE}}\big(Y^* - \hat{Y}^*\big) = \hat{\sigma}\sqrt{1 + (\underset{\sim}{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1}\underset{\sim}{x}^*}$$

$$\text{Var}\big(Y^* - \hat{Y}^*\big) = \sigma^2\big(1 + (\underset{\sim}{x}^*)^\top (\mathbf{X}'\mathbf{X})^{-1}\underset{\sim}{x}^*\big)$$

Let $\hat{\Sigma} \stackrel{\text{def}}{=} \widehat{\text{Var}}\big(\hat{\beta}\big) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$.

$$\widehat{\text{Var}}(\hat{\mu}(\underset{\sim}{x})) = \underset{\sim}{x}^\top \hat{\Sigma}\underset{\sim}{x}$$

Let $\Delta\mu(\underset{\sim}{x}, \underset{\sim}{x}^*) = \mu(\underset{\sim}{x}) - \mu(\underset{\sim}{x}^*)$, and let $\Delta\underset{\sim}{x} = \underset{\sim}{x} - \underset{\sim}{x}^*$; then:

$$\widehat{\text{Var}}\big(\widehat{\Delta\mu}(\underset{\sim}{x}, \underset{\sim}{x}^*)\big) = \Delta\underset{\sim}{x}^\top \hat{\Sigma}\Delta\underset{\sim}{x}$$

## Epi 204: Generalized linear models

Generalized linear models have three components:

1. The **outcome distribution** family: $\text{p}(Y|\mu(\underset{\sim}{x}))$
2. The **link function**: $g(\mu(\underset{\sim}{x})) = \eta(\underset{\sim}{x})$
3. The **linear component**: $\eta(\underset{\sim}{x}) = \underset{\sim}{x} \cdot \beta$

$$_\omega(\underset{\sim}{x}, \underset{\sim}{x}^*) = \exp\{(\Delta\underset{\sim}{x}) \cdot \underset{\sim}{\beta}\}$$

**Estimates of odds ratios from 2x2 contingency tables**

$$\hat{\theta} = \frac{ad}{bc}$$

**Summary of logistic regression definitions and results**

**Odds and log-odds**

These association-measure results are covered in Measures of Association for Binary Outcomes[1].

**Odds** (def[2])

$$\omega \stackrel{\text{def}}{=} \frac{\Pr(A)}{\Pr(\neg A)}$$

**Conditional odds** (def[3])

$$\omega(A|B) \stackrel{\text{def}}{=} \frac{\Pr(A|B)}{\Pr(\neg A|B)}$$

**Odds function** (def[4])

$$\text{odds}\{\pi\} \stackrel{\text{def}}{=} \frac{\pi}{1 - \pi}$$

[1] binary-outcome-associations.qmd
[2] binary-outcome-associations.qmd#def-odds
[3] binary-outcome-associations.qmd#def-c-odds
[4] binary-outcome-associations.qmd#def-odds-fn

**Probability to odds** (thm[5], cor[6])

$$\omega = \frac{\pi}{1-\pi} = \text{odds}\{\pi\}$$

**Simplified odds expressions; odds of a non-event** (thm[7], cor[8])

$$\text{odds}\{\pi\} = \frac{1}{\pi^{-1}-1} = (\pi^{-1}-1)^{-1}, \qquad \omega(\neg A) = \frac{1-\pi}{\pi} = \pi^{-1}-1$$

**Odds ratio** (def[9])

$$(\omega_1, \omega_2) \stackrel{\text{def}}{=} \frac{\omega_1}{\omega_2}$$

**OR as ratio of probability ratios** (thm[10])

$$(\omega_1, \omega_2) = \frac{\omega_1}{\omega_2} = \frac{\left(\frac{\pi_1}{1-\pi_1}\right)}{\left(\frac{\pi_2}{1-\pi_2}\right)}$$

**Odds ratios are reversible, marginally and conditionally** (thm[11], thm[12])

$$_{\omega}(A|B) = \ _{\omega}(B|A), \qquad _{\omega}(A|B,C) = \ _{\omega}(B|A,C)$$

**Inverse-odds and probability recovery**
**Odds to probability** (thm[13])

$$\pi = \frac{\omega}{1+\omega}$$

**Inverse-odds function** (def[14]; recovers probability by cor[15])

$$\text{invodds}\{\omega\} \stackrel{\text{def}}{=} \frac{\omega}{1+\omega}, \qquad \pi = \text{invodds}\{\omega\}$$

**Simplified inverse-odds** (lem[16])

$$\text{invodds}\{\omega\} = \frac{1}{1+\omega^{-1}} = (1+\omega^{-1})^{-1}$$

**One minus inverse-odds, and its complement** (lem[17], cor[18])

$$1-\pi = \frac{1}{1+\omega}, \qquad 1+\omega = \frac{1}{1-\pi}$$

**Log-odds (logit) and expit**
**Log-odds, and log-odds from probability** (def[19], thm[20])

$$\eta \stackrel{\text{def}}{=} \log\{\omega\}, \qquad \eta = \log\left\{\frac{\pi}{1-\pi}\right\}$$

**Logit function, and expanded form** (def[21], thm[22])

$$\text{logit}(\pi) \stackrel{\text{def}}{=} \log\{\text{odds}\{\pi\}\} = \log\left\{\frac{\pi}{1-\pi}\right\}$$

[5] binary-outcome-associations.qmd#thm-prob-to-odds
[6] binary-outcome-associations.qmd#cor-oddsf-to-odds
[7] binary-outcome-associations.qmd#thm-odds-simplified
[8] binary-outcome-associations.qmd#cor-odds-neg
[9] binary-outcome-associations.qmd#def-OR
[10] binary-outcome-associations.qmd#thm-or-ratio-ratio
[11] binary-outcome-associations.qmd#thm-or-swap
[12] binary-outcome-associations.qmd#thm-conditional-OR-swap
[13] binary-outcome-associations.qmd#thm-odds-to-prob
[14] binary-outcome-associations.qmd#def-inv-odds
[15] binary-outcome-associations.qmd#cor-invodds-pi
[16] binary-outcome-associations.qmd#lem-invodds-simplified
[17] binary-outcome-associations.qmd#lem-one-minus-odds-inv
[18] binary-outcome-associations.qmd#cor-inverse-odds-nonevent
[19] binary-outcome-associations.qmd#def-logodds
[20] binary-outcome-associations.qmd#thm-logodds-pi
[21] binary-outcome-associations.qmd#def-logit-fn
[22] binary-outcome-associations.qmd#thm-logit-function

**Log-odds equals logit** (cor[23])

$$\eta = \text{logit}\{\pi\}$$

**Odds and probability from log-odds** (lem[24], thm[25])

$$\omega = \exp\{\eta\}, \qquad \pi = \frac{\exp\{\eta\}}{1+\exp\{\eta\}}$$

**Expit / inverse-logit function, and equivalent expressions** (def[26], thm[27])

$$\text{expit}(\eta) \stackrel{\text{def}}{=} \text{invodds}\{\exp\{\eta\}\} = \frac{\exp\{\eta\}}{1+\exp\{\eta\}} = (1+\exp\{-\eta\})^{-1}$$

**Probability as expit** (thm[28])

$$\pi = \text{expit}\{\eta\} \tag{1}$$

**Logit and expit are inverses** (thm[29])

$$\text{logit}\{\text{expit}\{\eta\}\} = \eta \qquad \text{expit}\{\text{logit}\{\pi\}\} = \pi$$

$$\overbrace{\left[\pi \stackrel{\text{def}}{=} \Pr(Y=1|\underset{\sim}{X}=\underset{\sim}{x})\right] \underbrace{\xrightarrow[\frac{\omega}{1+\omega}]{\frac{\pi}{1-\pi}} \left[\omega \stackrel{\text{def}}{=} \text{odds}\{Y=1|\underset{\sim}{X}=\underset{\sim}{x}\}\right] \xrightarrow[\exp\{\eta\}]{\log\{\omega\}} \left[\eta(\underset{\sim}{x}) \stackrel{\text{def}}{=} \eta(Y=1|\underset{\sim}{X}=\underset{\sim}{x})\right]}}^{\text{logit}(\pi)}$$
$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{expit}(\eta)}$$

Figure 1: Diagram of logistic regression[30] link and inverse link functions[31]

**Rare events**

**Odds minus probability** (thm[32])

$$\omega - \pi = \frac{\pi^2}{1-\pi}, \quad \text{where } \omega = \frac{\pi}{1-\pi}$$

**OR as RR times a correction factor** (thm[33])

$$(\omega_1, \omega_2) = \rho(\pi_1, \pi_2) \cdot \frac{1-\pi_2}{1-\pi_1}$$

When $\pi_1$ and $\pi_2$ are both small, the correction factor is close to 1, so the odds ratio[34] is close to the risk ratio[35].

**Derivatives**

| | $\pi$ | $\omega$ | $\eta$ | $\underset{\sim}{x}$ | $\underset{\sim}{\beta}$ |
|---|---|---|---|---|---|
| $\pi$ | $1$ | $(1+\omega)^2$ | $\frac{(1+\omega)^2}{\omega}$ | undef | undef |
| $\omega$ | $(1-\pi)^2$ | $1$ | $\frac{1}{\omega}$ | undef | undef |
| $\eta$ | $\pi(1-\pi)$ | $\omega$ | $1$ | undef | undef |
| $\underset{\sim}{x}$ | $\underbrace{\underset{\sim}{\beta}\pi(1-\pi)}_{p\times 1}$ | $\underset{\sim}{\underset{\sim}{\beta}}\underset{\sim}{\omega}$ $_{p\times 1}$ | $\underset{\sim}{\underset{\sim}{\beta}}$ $_{p\times 1}$ | $\underset{\sim}{\underset{\sim}{\mathbb{I}}}$ $_{p\times p}$ | $\mathbf{0}_{p\times p}$ |
| $\underset{\sim}{\beta}$ | $\underbrace{\underset{\sim}{x}\pi(1-\pi)}_{p\times 1}$ | $\underset{\sim}{\underset{\sim}{x}}\underset{\sim}{\omega}$ $_{p\times 1}$ | $\underset{\sim}{\underset{\sim}{x}}$ $_{p\times 1}$ | $\mathbf{0}_{p\times p}$ | $\underset{\sim}{\underset{\sim}{\mathbb{I}}}$ $_{p\times p}$ |

[23] binary-outcome-associations.qmd#cor-logodds-logit
[24] binary-outcome-associations.qmd#lem-odds-from-logodds
[25] binary-outcome-associations.qmd#thm-prob-from-logodds
[26] binary-outcome-associations.qmd#def-expit
[27] binary-outcome-associations.qmd#thm-expit-expressions
[28] binary-outcome-associations.qmd#thm-expit-prob-logodds
[29] binary-outcome-associations.qmd#thm-logit-expit
[32] binary-outcome-associations.qmd#thm-odds-minus-probs
[33] binary-outcome-associations.qmd#thm-rare-disease-or-rr
[34] binary-outcome-associations.qmd#def-OR
[35] binary-outcome-associations.qmd#def-RR

Column labels indicate the numerators of the derivatives[36]; row labels indicate the denominators (dimensions and details in tbl[37]).

**Derivative of expit** (cor[38])

$$\text{expit}'\{\eta\} = \text{expit}\{\eta\}(1 - \text{expit}\{\eta\})$$

**Logistic regression model**

**Logistic regression** (def[39])

$$Y_i | \underset{\sim}{X}_i \ \sim_{\perp} \ \text{Ber}(\pi(\underset{\sim}{X}_i)), \qquad \text{logit}\{\pi(\underset{\sim}{x})\} = \underset{\sim}{x}^\top \underset{\sim}{\beta}$$

Throughout this summary, $\underset{\sim}{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)$ is the $(p+1) \times 1$ coefficient vector (including the intercept), $\mathbf{X}$ is the $n \times (p+1)$ design matrix whose $i^{\text{th}}$ row is $\underset{\sim}{x}_i{}^\top$, and $\underset{\sim}{\varepsilon}$ is the $n \times 1$ error vector with components $\varepsilon_i \overset{\text{def}}{=} y_i - \pi_i$.

**Log-likelihood and score function**

**Log-likelihood component** (lem[40])

$$\ell_i(\pi_i) = y_i \eta_i - \log\{1 + \omega_i\}$$

**Score function as sum of components** (lem[41], thm[42])

$$\underset{\sim}{\ell}'(\underset{\sim}{\beta}) = \sum_{i=1}^{n} \underset{\sim}{\ell}_i'(\underset{\sim}{\beta}), \qquad \underset{\sim}{\ell}_i'(\underset{\sim}{\beta}) = \underset{\sim}{x}_i \varepsilon_i$$

**Score function**[43] (thm[44])

$$\underset{\sim}{\ell}'(\underset{\sim}{\beta}) = \sum_{i=1}^{n} \underset{\sim}{x}_i \varepsilon_i = \underset{(p+1)\times n}{\underset{\sim}{\mathbf{X}}^\top} \underset{n\times 1}{\underset{\sim}{\varepsilon}}$$

**Maximum likelihood estimation**

**Estimating equation** (eq[45]) The MLE[46] $\underset{\sim}{\hat{\beta}}$ solves the score equation[47]:

$$\underbrace{\underset{\sim}{\ell}'(\underset{\sim}{\hat{\beta}})}_{(p+1)\times 1} = \sum_{i=1}^{n} \underbrace{\underset{\sim}{x}_i}_{(p+1)\times 1} \underbrace{\left(y_i - \text{expit}\left\{\underset{\sim}{x}_i{}^\top \underset{\sim}{\hat{\beta}}\right\}\right)}_{1\times 1} = \mathbf{0}_{(p+1)\times 1}$$

This estimating equation has no closed-form solution in general; $\underset{\sim}{\hat{\beta}}$ must be computed by numerical iteration.

**Hessian** (eq[48])

$$\underbrace{\underset{\sim}{\ell}''(\underset{\sim}{\beta})}_{(p+1)\times(p+1)} = - \underset{(p+1)\times n}{\underset{\sim}{\mathbf{X}}^\top} \underset{n\times n}{\underset{\sim}{\mathbf{D}}} \underset{n\times(p+1)}{\underset{\sim}{\mathbf{X}}}, \qquad \mathbf{D} \overset{\text{def}}{=} \text{diag}(\text{Var}(Y_i | \underset{\sim}{X}_i = \underset{\sim}{x}_i)) = \text{diag}(\pi_i(1 - \pi_i))$$

**Concavity** (rem[49]): the Hessian[50] is negative semi-definite for every $\beta$ (negative definite when $\mathbf{X}$ has full column rank), so the log-likelihood[51] is concave[52] and every solution of the score equation is a global maximum.

**Newton-Raphson update** (Newton-Raphson method[53])

$$\underbrace{\underset{\sim}{\hat{\beta}}^*}_{(p+1)\times 1} \leftarrow \underbrace{\underset{\sim}{\hat{\beta}}^*}_{(p+1)\times 1} - \underbrace{\left(\underset{\sim}{\ell}''\left(\underset{\sim}{y}; \underset{\sim}{\hat{\beta}}^*\right)\right)^{-1}}_{(p+1)\times(p+1)} \underbrace{\underset{\sim}{\ell}'\left(\underset{\sim}{y}; \underset{\sim}{\hat{\beta}}^*\right)}_{(p+1)\times 1}$$

**One-sample MLE for odds** (thm[54])

$$\hat{\omega} = \frac{x}{n - x}$$

[36] https://morrison-lab.github.io/mds/calculus-derivatives.html#def-derivative
[37] logistic-regression.qmd#tbl-logistic-deriv-matrix
[38] logistic-regression.qmd#cor-deriv-expit
[39] logistic-regression.qmd#def-logistic-regression
[40] logistic-regression.qmd#lem-logistic-loglik-component
[41] logistic-regression.qmd#lem-score-logistic
[42] logistic-regression.qmd#thm-logistic-score-comp
[43] https://morrison-lab.github.io/sds/intro-MLEs.html#def-score
[44] logistic-regression.qmd#thm-logistic-score-fn
[45] logistic-regression.qmd#eq-score-logistic
[46] https://morrison-lab.github.io/sds/intro-MLEs.html#def-mle
[47] https://morrison-lab.github.io/sds/intro-MLEs.html#def-score-equation
[48] logistic-regression.qmd#eq-logistic-hess
[49] logistic-regression.qmd#rem-hessian-nsd
[50] https://morrison-lab.github.io/mds/vector-calculus.html#def-hessian
[51] https://morrison-lab.github.io/sds/intro-MLEs.html#def-loglik
[52] https://morrison-lab.github.io/mds/optimization.html#def-strictly-convex
[53] https://morrison-lab.github.io/sds/intro-MLEs.html#sec-newton-raphson
[54] binary-outcome-associations.qmd#thm-est-odds

**Wald tests and confidence intervals for coefficients**

**Wald test statistic** for $H_0 : \beta_k = \beta_{k,0}$ (CLT for MLEs[55])

$$z_k = \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\mathrm{SE}}\big(\hat{\beta}_k\big)}, \qquad z_k \dot{\sim} N(0,1) \text{ under } H_0$$

**95% confidence intervals for $\beta_k$ and $e^{\beta_k}$** (MLE[56] invariance)

$$\hat{\beta}_k \pm 1.96 \cdot \widehat{\mathrm{SE}}\big(\hat{\beta}_k\big), \qquad e^{\beta_k} \in \Big(e^{\hat{\beta}_k - 1.96 \cdot \widehat{\mathrm{SE}}(\hat{\beta}_k)},\ e^{\hat{\beta}_k + 1.96 \cdot \widehat{\mathrm{SE}}(\hat{\beta}_k)}\Big)$$

For $k \neq 0$, $e^{\beta_k}$ is an odds ratio[57]; for $k = 0$, $e^{\beta_0}$ is the baseline odds[58].

**Odds ratios in logistic regression**

**General OR formula** (lem[59])

$$\mathrm{OR}(\underset{\sim}{x}, \underset{\sim}{x}^*) = \exp\{\eta(\underset{\sim}{x}) - \eta(\underset{\sim}{x}^*)\}$$

**Difference in log-odds; OR in terms of $\Delta\eta$** (def[60], cor[61])

$$\Delta\eta \overset{\text{def}}{=} \eta(\underset{\sim}{x}) - \eta(\underset{\sim}{x}^*), \qquad \mathrm{OR}(\underset{\sim}{x}, \underset{\sim}{x}^*) = \exp\{\Delta\eta\}$$

**$\Delta\eta$ from covariates** (lem[62])

$$\Delta\eta = (\underset{\sim}{x} - \underset{\sim}{x}^*) \cdot \underset{\sim}{\beta}$$

**Difference in covariate patterns; $\Delta\eta$ from $\Delta\underset{\sim}{x}$** (def[63], cor[64])

$$\Delta\underset{\sim}{x} \overset{\text{def}}{=} \underset{\sim}{x} - \underset{\sim}{x}^*, \qquad \Delta\eta = \Delta\underset{\sim}{x} \cdot \underset{\sim}{\beta}$$

**OR in terms of $\Delta\underset{\sim}{x}$** (thm[65])

$$\mathrm{OR}(\underset{\sim}{x}, \underset{\sim}{x}^*) = \exp\{(\Delta\underset{\sim}{x}) \cdot \underset{\sim}{\beta}\}$$

**Log OR equals $\Delta\eta$** (cor[66])

$$\log\{\mathrm{OR}(\underset{\sim}{x}, \underset{\sim}{x}^*)\} = \Delta\eta$$

**Inference for log-odds and odds ratios**

**Estimated SE of log-odds** (thm[67])

$$\widehat{\mathrm{Var}}(\hat{\eta}(\underset{\sim}{x})) = \underset{1\times(p+1)}{\underset{\sim}{x}^\top} \underset{(p+1)\times(p+1)}{\hat{\underset{\sim}{\Sigma}}} \underset{(p+1)\times 1}{\underset{\sim}{x}}, \qquad \widehat{\mathrm{SE}}(\hat{\eta}(\underset{\sim}{x})) = \sqrt{\underset{\sim}{x}^\top \hat{\underset{\sim}{\Sigma}} \underset{\sim}{x}}$$

**Estimated SE of $\Delta\hat{\eta}$** (thm[68])

$$\widehat{\mathrm{Var}}(\Delta\hat{\eta}) = \underset{1\times(p+1)}{\underset{\sim}{\Delta x}^\top} \underset{(p+1)\times(p+1)}{\hat{\underset{\sim}{\Sigma}}} \underset{(p+1)\times 1}{\underset{\sim}{(\Delta x)}}, \qquad \widehat{\mathrm{SE}}(\Delta\hat{\eta}) = \sqrt{\underset{\sim}{\Delta x}^\top \hat{\underset{\sim}{\Sigma}} \underset{\sim}{(\Delta x)}}$$

**CI for an odds ratio** (eq[69]; details in sec[70]): exponentiate the endpoints of the CI for $\Delta\eta$:

$$\mathrm{OR}(\underset{\sim}{x}, \underset{\sim}{x}^*) \in (e^L, e^R), \qquad (L, R) = \widehat{\Delta\eta} \mp 1.96 \cdot \widehat{\mathrm{SE}}\big(\widehat{\Delta\eta}\big)$$

[55] https://morrison-lab.github.io/sds/intro-MLEs.html#thm-dist-mle
[56] https://morrison-lab.github.io/sds/intro-MLEs.html#def-mle
[57] binary-outcome-associations.qmd#def-OR
[58] binary-outcome-associations.qmd#def-odds
[59] logistic-regression.qmd#lem-logistic-OR
[60] logistic-regression.qmd#def-difflogodds
[61] logistic-regression.qmd#cor-logistic-OR
[62] logistic-regression.qmd#lem-diff-logodds
[63] logistic-regression.qmd#def-diff-covs
[64] logistic-regression.qmd#cor-diff-logodds
[65] logistic-regression.qmd#thm-logistic-OR
[66] logistic-regression.qmd#cor-log-or
[67] logistic-regression.qmd#thm-est-se-logodds
[68] logistic-regression.qmd#thm-est-se-diff-logodds
[69] logistic-regression.qmd#eq-ci-OR-exp
[70] logistic-regression.qmd#sec-or-inference

**Odds ratios in study designs**

**Disease and exposure odds ratios** (def[71], def[72])

$$_{\omega}(D|E) \stackrel{\text{def}}{=} \frac{\omega(D|E)}{\omega(D|\neg E)}, \qquad _{\omega}(E|D) \stackrel{\text{def}}{=} \frac{\omega(E|D)}{\omega(E|\neg D)}$$

**Case-control studies** (section[73]): outcome-stratified sampling breaks estimation of risks[74], risk ratios[75], and the components of the disease odds ratio (def[76]), but the exposure odds ratio (def[77]) is estimable, and it equals the disease odds ratio by reversibility (thm[78]).

$2 \times 2$ **table shortcut** (table[79]; cell counts $a, b$ = exposed events and non-events, $c, d$ = unexposed events and non-events):

$$\hat{\omega}(Event|Exposed) = \frac{a}{b}, \qquad \hat{\omega}(Event|\neg Exposed) = \frac{c}{d}, \qquad _{\omega}(Exposed, \neg Exposed) = \frac{ad}{bc}$$

**Model fit and comparison**

Details in sec[80]; here $q$ is the number of distinct covariate patterns, $p$ is the number of $\beta$ parameters, and $\ell_{\text{full}}$ is the saturated-model log-likelihood.

**Deviance test** (def[81])

$$D \stackrel{\text{def}}{=} 2(\ell_{\text{full}} - \ell(\hat{\beta}; \mathbf{x})), \qquad D \mathrel{\dot{\sim}} \chi^2(q-p)$$

**Hosmer-Lemeshow test** (def[82]; $g$ groups by predicted probability)

$$X^2 \stackrel{\text{def}}{=} \sum_{c=1}^{g} \frac{(o_c - e_c)^2}{e_c} \mathrel{\dot{\sim}} \chi^2(g-2)$$

**Pearson chi-squared GOF test** (sum of squared Pearson residuals, def[83])

$$X^2 = \sum_{k=1}^{q} X_k^2 \mathrel{\dot{\sim}} \chi^2(q-p)$$

**AIC and BIC** [lower is better] (AIC[84] [1]; BIC[85] [2])

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p, \qquad \text{BIC} = -2\ell(\hat{\theta}) + p\log\{n\}$$

**Likelihood ratio test** (def[86]; for nested models with $p_0 < p_1$ parameters — here $\ell_1$ and $\ell_0$ are the two fitted models' log-likelihoods[87], not the saturated model[88]'s)

$$\Lambda \stackrel{\text{def}}{=} 2(\ell_1(\hat{\theta}_1) - \ell_0(\hat{\theta}_0)), \qquad \Lambda \mathrel{\dot{\sim}} \chi^2(p_1 - p_0) \text{ under } H_0$$

by Wilks' theorem[89].

**Residual-based diagnostics**

Residuals are computed per covariate pattern $k$ (grouped data); $h_k$ denotes the leverage of pattern $k$ (def[90]).

**Response residual** (def[91])

$$e_k \stackrel{\text{def}}{=} \bar{y}_k - \hat{\pi}(x_k)$$

[71] binary-outcome-associations.qmd#def-disease-OR
[72] binary-outcome-associations.qmd#def-exposure-OR
[73] binary-outcome-associations.qmd#sec-CC-studies
[74] binary-outcome-associations.qmd#def-risk-epi
[75] binary-outcome-associations.qmd#def-RR
[76] binary-outcome-associations.qmd#def-disease-OR
[77] binary-outcome-associations.qmd#def-exposure-OR
[78] binary-outcome-associations.qmd#thm-or-swap
[79] binary-outcome-associations.qmd#tbl-2x2-generic
[80] logistic-regression.qmd#sec-gof
[81] logistic-regression.qmd#def-deviance-test
[82] logistic-regression.qmd#def-hosmer-lemeshow
[83] logistic-regression.qmd#def-pearson-residual
[84] Linear-models-overview.qmd#def-aic
[85] Linear-models-overview.qmd#def-bic
[86] logistic-regression.qmd#def-lrt
[87] https://morrison-lab.github.io/sds/intro-MLEs.html#def-loglik
[88] Linear-models-overview.qmd#def-saturated-model
[89] https://morrison-lab.github.io/sds/intro-MLEs.html#thm-wilks
[90] logistic-regression.qmd#def-leverage-glm
[91] logistic-regression.qmd#def-response-residual-logistic

**Pearson residual (and standardized version)** (def[92], def[93])

$$X_k \overset{\text{def}}{=} \frac{\bar{y}_k - \hat{\pi}_k}{\sqrt{\hat{\pi}_k(1-\hat{\pi}_k)/n_k}}, \qquad r_{P_k} \overset{\text{def}}{=} \frac{X_k}{\sqrt{1-h_k}}$$

**Deviance residual (and standardized version)** (def[94])

$$d_k \overset{\text{def}}{=} \text{sign}(y_k - n_k\hat{\pi}_k)\left\{\sqrt{2[\ell_{\text{full}}(x_k) - \ell(\hat{\beta}; x_k)]}\right\}, \qquad r_{D_k} \overset{\text{def}}{=} \frac{d_k}{\sqrt{1-h_k}}$$

**Cook's distance** (def[95])

$$D_k \overset{\text{def}}{=} \frac{(r_{P_k})^2}{p} \cdot \frac{h_k}{1-h_k}$$

**Leverage** (def[96]): $h_k$ is the $k$-th diagonal element of the weighted hat matrix[97] (with $\mathbf{X}$ here the $q \times p$ design matrix[98] of covariate patterns):

$$\underset{q\times q}{\mathbf{H}} \overset{\text{def}}{=} \underbrace{\mathbf{W}^{1/2}}_{q\times q}\ \underset{q\times p}{\mathbf{X}}\left(\underbrace{\mathbf{X}^\top\mathbf{W}\mathbf{X}}_{p\times p}\right)^{-1}\underbrace{\mathbf{X}^\top}_{p\times q}\ \underbrace{\mathbf{W}^{1/2}}_{q\times q}, \qquad \underset{q\times q}{\mathbf{W}} \overset{\text{def}}{=} \text{diag}(n_k\,\hat{\pi}_k\,(1-\hat{\pi}_k))$$

**Comparing probabilities**

**Risk difference** (def[99])

$$\delta(\pi_1, \pi_2) \overset{\text{def}}{=} \pi_1 - \pi_2$$

**Risk ratio** (def[100])

$$\rho(\pi_1, \pi_2) \overset{\text{def}}{=} \frac{\pi_1}{\pi_2}$$

**Relative risk difference** (def[101]; equals RR minus 1 by thm[102])

$$\xi(\pi_1, \pi_2) \overset{\text{def}}{=} \frac{\delta(\pi_1, \pi_2)}{\pi_2} = \rho(\pi_1, \pi_2) - 1$$

**Marginal risks from logistic regression**

**Conditional predicted risk** (def[103])

$$\pi(a, z) \overset{\text{def}}{=} \Pr(Y = 1 \mid A = a, Z = z)$$

**Conditional RD and RR** (def[104], def[105])

$$\text{RD}(z) \overset{\text{def}}{=} \pi(1, z) - \pi(0, z), \qquad \text{RR}(z) \overset{\text{def}}{=} \frac{\pi(1, z)}{\pi(0, z)}$$

**Observed marginal risk** (def[106]; standardization form by thm[107])

$$\pi(a) \overset{\text{def}}{=} \Pr(Y = 1 \mid A = a) = \mathbb{E}[\pi(a, Z) \mid A = a]$$

**Potential mean and causal marginal risk** (def[108], def[109]; identification form by thm[110])

$$\mu_a \overset{\text{def}}{=} \mathbb{E}[Y^a], \qquad \pi_a \overset{\text{def}}{=} \Pr(Y^a = 1) = \mathbb{E}[\pi(a, Z)]$$

[92] logistic-regression.qmd#def-pearson-residual
[93] logistic-regression.qmd#def-standardized-pearson-residual
[94] logistic-regression.qmd#def-deviance-residual-logistic
[95] logistic-regression.qmd#def-cooks-distance-glm
[96] logistic-regression.qmd#def-leverage-glm
[97] Linear-models-overview.qmd#def-hat-matrix
[98] https://morrison-lab.github.io/mds/linear-algebra-projections.html#def-design-matrix
[99] binary-outcome-associations.qmd#def-RD
[100] binary-outcome-associations.qmd#def-RR
[101] binary-outcome-associations.qmd#def-RRD
[102] binary-outcome-associations.qmd#thm-rrd-vs-rr
[103] logistic-regression.qmd#def-cond-predicted-risk
[104] logistic-regression.qmd#def-cond-rd
[105] logistic-regression.qmd#def-cond-rrr
[106] logistic-regression.qmd#def-observed-marginal-risk
[107] logistic-regression.qmd#thm-observed-marginal-risk
[108] logistic-regression.qmd#def-potential-mean
[109] logistic-regression.qmd#def-causal-marginal-risk
[110] logistic-regression.qmd#thm-causal-marginal-risk

**No confounding: observed = causal** (cor[111]; the two estimands[112] are contrasted in rem[113])

$$Z \perp\!\!\!\perp A \implies \pi(a) = \pi_a$$

**G-computation estimator** (def[114]; consistent for $\pi_a$ by thm[115])

$$\hat{\pi}_a \overset{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} \hat{\pi}_{a,i}, \qquad \hat{\pi}_{a,i} \overset{\text{def}}{=} \hat{\pi}(a, z_i)$$

**Predictive margins** (def[116]; full-sample form def[117] is consistent for $\pi_a$ by cor[118])

$$\widehat{\text{PM}}(a \mid S) \overset{\text{def}}{=} \frac{1}{|S|}\sum_{i \in S} \hat{\pi}_{a,i}, \qquad \widehat{\text{PM}}(a) \overset{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} \hat{\pi}_{a,i} = \hat{\pi}_a$$

**Exposed-subgroup predictive margin** (def[119]; consistent for $\pi(a)$ by thm[120]; equals $\bar{Y}_a$ exactly by lem[121])

$$\hat{\pi}_a(a) \overset{\text{def}}{=} \frac{1}{n_a}\sum_{i:A_i=a} \hat{\pi}_{a,i} = \bar{Y}_a$$

**Observed marginal RD and RR** (def[122], def[123])

$$\text{OMRD} \overset{\text{def}}{=} \pi(1) - \pi(0), \qquad \text{OMRR} \overset{\text{def}}{=} \frac{\pi(1)}{\pi(0)}$$

**Causal marginal RD and RR** (def[124], def[125])

$$\text{CMRD} \overset{\text{def}}{=} \pi_1 - \pi_0, \qquad \text{CMRR} \overset{\text{def}}{=} \frac{\pi_1}{\pi_0}$$

**Bootstrap SE and CI for marginal contrasts** (def[126]; see Bootstrap Confidence Intervals[127] and exm[128]): resample the data with replacement $B$ times; refit the logistic model and recompute the marginal contrast on each resample; the bootstrap SE is the standard deviation[129] of the $B$ estimates, and a 95% percentile CI uses their 2.5th and 97.5th percentiles.

**Collapsibility** (def[130]): an estimand $\theta$ is collapsible with respect to $Z$ if

$$\theta \overset{\text{def}}{=} \mathbb{E}[\theta(Z)]$$

Risk differences[131] are collapsible; risk ratios[132] and odds ratios[133] are not collapsible in general (thm[134]; counterexample in exm[135]).

## Survival analysis

### Probability distribution functions

[111] logistic-regression.qmd#cor-observed-equals-causal
[112] https://morrison-lab.github.io/sds/estimation.html#def-estimand
[113] logistic-regression.qmd#rem-observed-vs-causal-marginal-risk
[114] logistic-regression.qmd#def-g-computation
[115] logistic-regression.qmd#thm-g-computation-consistency
[116] logistic-regression.qmd#def-predictive-margins
[117] logistic-regression.qmd#def-full-sample-predictive-margin
[118] logistic-regression.qmd#cor-full-sample-pm
[119] logistic-regression.qmd#def-subgroup-predictive-margin
[120] logistic-regression.qmd#thm-subgroup-pm
[121] logistic-regression.qmd#lem-subgroup-pm-proportion
[122] logistic-regression.qmd#def-observed-marginal-rd
[123] logistic-regression.qmd#def-observed-marginal-rr
[124] logistic-regression.qmd#def-causal-marginal-rd
[125] logistic-regression.qmd#def-causal-marginal-rr
[126] logistic-regression.qmd#def-bootstrap-marginal-rd
[127] https://morrison-lab.github.io/sds/bootstrap.html#sec-bootstrap-ci
[128] logistic-regression.qmd#exm-wcgs-marginal-rd
[129] https://morrison-lab.github.io/pds/variance-covariance.html#def-sd
[130] logistic-regression.qmd#def-collapsibility
[131] binary-outcome-associations.qmd#def-RD
[132] binary-outcome-associations.qmd#def-RR
[133] binary-outcome-associations.qmd#def-OR
[134] logistic-regression.qmd#thm-collapsibility
[135] logistic-regression.qmd#exm-collapsibility

Table 1: Probability distribution functions

| Name | Symbols | Definition |
|---|---|---|
| Probability density function (PDF)[136] | $\mathrm{f}(t), \mathrm{p}(t)$ | $\mathrm{p}(T = t)$ |
| Cumulative distribution function (CDF)[137] | $\mathrm{F}(t), \mathbb{P}(t)$ | $\mathbb{P}(T \leq t)$ |
| Survival function[138] | $\mathrm{S}(t), \bar{\mathrm{F}}(t)$ | $\mathbb{P}(T > t)$ |
| Hazard function[139] | $\lambda(t), \mathrm{h}(t)$ | $\mathrm{p}(T = t \vert T \geq t)$ |
| Cumulative hazard function[140] | $\Lambda(t), \mathrm{H}(t)$ | $\int_{u=-\infty}^{t} \lambda(u)du$ |
| Log-hazard function[141] | $\eta(t)$ | $\log\{\lambda(t)\}$ |

**Diagram of survival distribution function relationships**

$$\mathrm{f}(t) \underset{\mathrm{S}(t)\lambda(t)}{\overset{-S'(t)}{\longleftarrow}} \mathrm{S}(t) \xleftarrow{\exp\{-\Lambda(t)\}} \Lambda(t) \overset{\int_{u=0}^{t}\lambda(u)du}{\longleftarrow} \lambda(t) \xleftarrow{\exp\{\eta(t)\}} \eta(t)$$

$$\mathrm{f}(t) \xrightarrow[\int_{u=t}^{\infty}\mathrm{f}(u)du]{\mathrm{f}(t)/\lambda(t)} \mathrm{S}(t) \xrightarrow[-\log \mathrm{S}(t)]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t)$$

**Survival likelihood contributions, assuming non-informative censoring**

$$\begin{aligned}\mathrm{p}(Y = y, D = d) &= [\mathrm{f}_T(y)]^d[\mathrm{S}_T(y)]^{1-d} \\ &= [\lambda_T(y)]^d[\mathrm{S}_T(y)]\end{aligned}$$

**Nonparametric time-to-event distribution estimators**

$$\hat{\lambda}_i = \frac{d_i}{r_i}$$

$$\hat{\kappa}_i = 1 - \hat{\lambda}_i = \frac{r_i - d_i}{r_i}$$

$$\hat{\mathrm{S}}_{\mathrm{KM}}(t) \overset{\text{def}}{=} \prod_{\{i:\ t_i \leq t\}} \hat{\kappa}_i$$

$$\hat{\Lambda}_{NA}(t) \overset{\text{def}}{=} \sum_{\{i:\ t_i \leq t\}} \hat{\lambda}_i$$

**Proportional hazards model structure**

| Formula | Description |
|---|---|
| $\lambda(t \mid \underset{\sim}{x}) = \lambda_0(t) \cdot \theta_\lambda(\underset{\sim}{x})$ | Proportional hazards assumption[142] |
| $\Lambda(t \mid \underset{\sim}{x}) = \Lambda_0(t) \cdot \theta_\lambda(\underset{\sim}{x})$ | Cumulative hazard factorization[143] |
| $\eta(t \mid \underset{\sim}{x}) = \eta_0(t) + \Delta\eta(\underset{\sim}{x})$ | Log-hazard decomposition[144] |
| $\Delta\eta(\underset{\sim}{x}) = \underset{\sim}{x} \cdot \beta = \beta_1 x_1 + \cdots + \beta_p x_p$ | Linear predictor[145] |
| $\theta_\lambda(\underset{\sim}{x}) = \exp\{\Delta\eta(\underset{\sim}{x})\}$ | Hazard multiplier[146] |
| $_\lambda(t \vert \underset{\sim}{x} : \underset{\sim}{x}^*) \overset{\text{def}}{=} \frac{\lambda(t\vert\underset{\sim}{x})}{\lambda(t\vert\underset{\sim}{x}^*)}$ | Hazard ratio definition[147] |
| $_\lambda(t \mid \underset{\sim}{x} : \underset{\sim}{x}^*) = \exp\{\Delta\eta(\underset{\sim}{x}) - \Delta\eta(\underset{\sim}{x}^*)\} = \exp\{(\underset{\sim}{x} - \underset{\sim}{x}^*) \cdot \underset{\sim}{\beta}\}$ | Hazard ratio formula[148] |

[136] https://morrison-lab.github.io/pds/random-variables.html#def-pdf
[137] https://morrison-lab.github.io/pds/random-variables.html#def-cdf
[138] intro-to-survival-analysis.qmd#def-surv-fn
[139] intro-to-survival-analysis.qmd#def-hazard
[140] intro-to-survival-analysis.qmd#def-cuhaz
[141] proportional-hazards-models.qmd#def-loghaz
[142] proportional-hazards-models.qmd#def-pha
[143] proportional-hazards-models.qmd#thm-ph-cuhaz
[144] proportional-hazards-models.qmd#thm-loghaz-decomp
[145] Intro-to-GLMs.qmd#def-linear-predictor
[146] proportional-hazards-models.qmd#def-risk-score
[147] intro-to-survival-analysis.qmd#def-hazard-ratio
[148] proportional-hazards-models.qmd#thm-hazard-ratio-ph

**Proportional hazards model partial likelihood formula:**

**Definition 0.1** (Cox PH partial likelihood)**.**

$$\mathcal{L}_i^*(\underset{\sim}{\beta}) \stackrel{\text{def}}{=} \frac{\theta_\lambda(\underset{\sim}{x}_{(i)})}{\sum_{k \in R(t_i)} \theta_\lambda(\underset{\sim}{x}_k)}$$
$$\mathcal{L}^*(\underset{\sim}{\beta}) \stackrel{\text{def}}{=} \prod_{\{i:\ d_i=1\}} \mathcal{L}_i^*(\underset{\sim}{\beta})$$

where, for the distinct event times $t_1 < \cdots < t_K$ of the data set:

- $(i)$ is the subject who fails at $t_i$ (unique under the no-ties assumption);
- $R(t_i) \stackrel{\text{def}}{=} \{j : Y_j \geq t_i\}$ is the risk set[a] at $t_i$ (where $Y_j = \min(T_j, C_j)$ is subject $j$'s observed follow-up time) — the subjects still under observation just before $t_i$;
- $d_i$ is the number of events at $t_i$ (so $d_i = 1$ when there are no ties);
- $\theta_\lambda(\underset{\sim}{x}) \stackrel{\text{def}}{=} \exp\{\underset{\sim}{x}^\top \underset{\sim}{\beta}\}$ is the hazard ratio (risk score[b]) for covariate pattern $\underset{\sim}{x}$.

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] proportional-hazards-models.qmd#def-risk-score

**Proportional hazards model baseline cumulative hazard estimator:**

**Definition 0.2** (Breslow estimator of the baseline cumulative hazard)**.**

$$\hat{\Lambda}_0(t) \stackrel{\text{def}}{=} \sum_{t_i \leq t} \frac{d_i}{\sum_{k \in R(t_i)} \theta_\lambda(\underset{\sim}{x}_k)}$$

where, for the distinct event times $t_1 < \cdots < t_K$ of the data set:

- $d_i$ is the number of events at $t_i$;
- $R(t_i) \stackrel{\text{def}}{=} \{j : Y_j \geq t_i\}$ is the risk set[a] at $t_i$ (where $Y_j = \min(T_j, C_j)$ is subject $j$'s observed follow-up time) — the subjects still under observation just before $t_i$;
- $\theta_\lambda(\underset{\sim}{x}) \stackrel{\text{def}}{=} \exp\{\underset{\sim}{x}^\top \underset{\sim}{\beta}\}$ is the hazard ratio (risk score[b]) for covariate pattern $\underset{\sim}{x}$.

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] proportional-hazards-models.qmd#def-risk-score

**Example 0.1** (Breslow estimator with two event times)**.** Suppose there are two distinct event times $t_1 = 2$ and $t_2 = 4$ (no ties, $d_1 = d_2 = 1$), with risk-set-weighted totals $\sum_{k \in R(2)} \theta_\lambda(\underset{\sim}{x}_k) = 10$ and $\sum_{k \in R(4)} \theta_\lambda(\underset{\sim}{x}_k) = 6$. Then

$$\hat{\Lambda}_0(4) = \frac{1}{10} + \frac{1}{6} \approx 0.267.$$


[1] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, 1974, Available: https://doi.org/10.1109/TAC.1974.1100705

[2] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978, Available: https://doi.org/10.1214/aos/1176344136